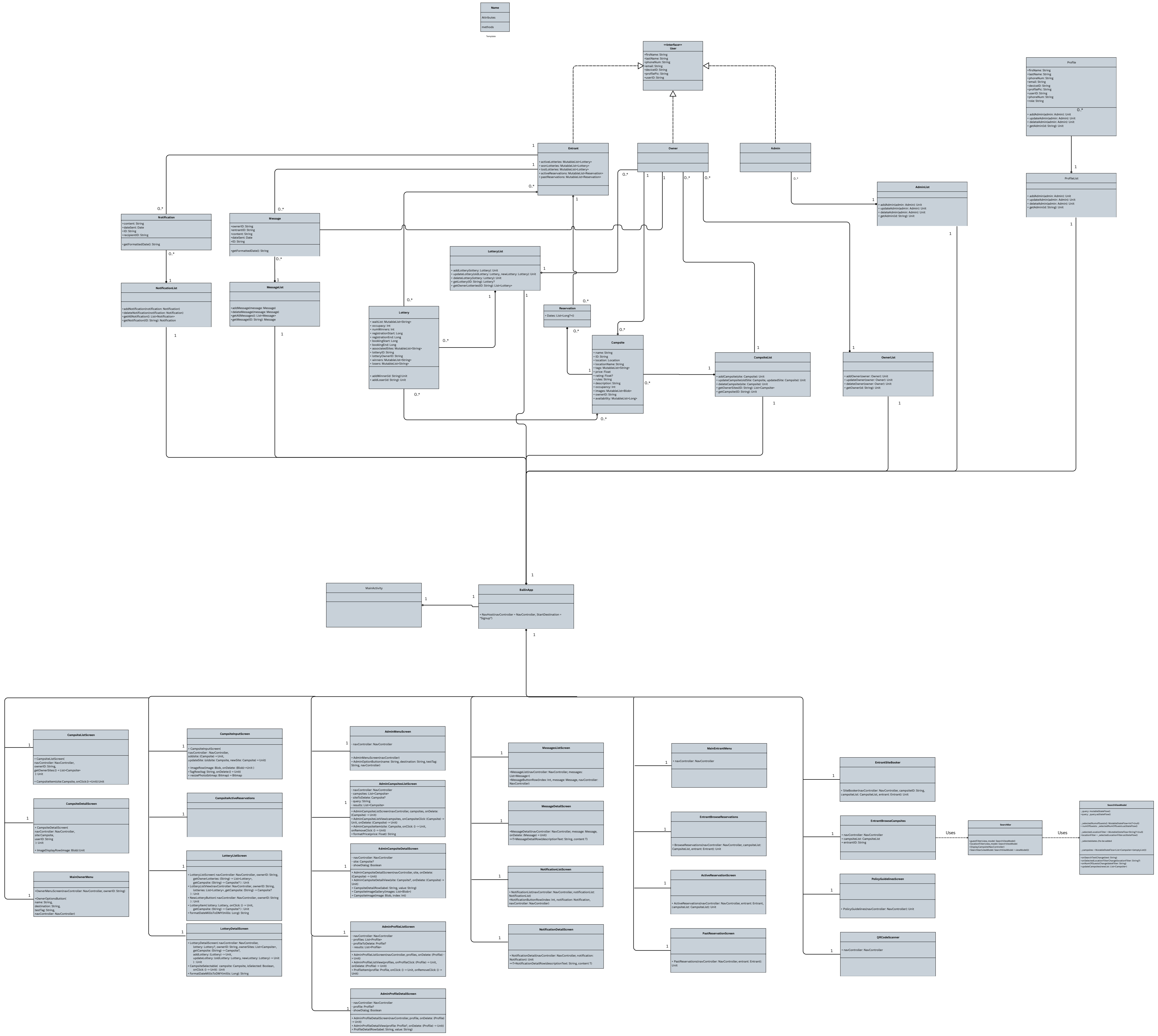
Project Ballin UML Class Diagram


Name
Attributes
methods
Template
<<interface>>
User
+firsName: String
+lastName: String
+phoneNum: String
+email: String
+deviceID: String
+profilePic: String
+userID: String
Entrant
+ activeLotteries: MutableList<Lottery>
+ wonLotteries: MutableList<Lottery>
+ lostLotteries: MutableList<Lottery>
+ activeReservations: MutableList<Reservation>
+ pastReservations: MutableList<Reservation>
Owner
Admin
Profile
+firsName: String
+lastName: String
+phoneNum: String
+email: String
+deviceID: String
+profilePic: String
+userID: String
+phoneNum: String
+role: String
+ addAdmin(admin: Admin): Unit
+ updateAdmin(admin: Admin): Unit
+ deleteAdmin(admin: Admin): Unit
+ getAdmin(id: String): Unit
ProfileList
+ addAdmin(admin: Admin): Unit
+ updateAdmin(admin: Admin): Unit
+ deleteAdmin(admin: Admin): Unit
+ getAdmin(id: String): Unit
Notification
+content: String
+dateSent: Date
+ID: String
+recipientID: String
+getFormattedDate(): String
Message
+ownerID: String
+entrantID: String
+content: String
+dateSent: Date
+ID: String
+getFormattedDate(): String
NotificationList
+addNotification(notification: Notification)
+deleteNotification(notification: Notification)
+getAllNotification(): List<Notification>
+getNotification(ID: String): Notification
MessageList
+addMessage(message: Message)
+deleteMessage(message: Message)
+getAllMessages(): List<Message>
+getMessage(ID: String): Message
LotteryList
+ addLottery(lottery: Lottery): Unit
+ updateLottery(oldLottery: Lottery, newLottery: Lottery): Unit
+ deleteLottery(lottery: Lottery): Unit
+ getLottery(ID: String): Lottery?
+ getOwnerLotteries(ID: String): List<Lottery>
Lottery
+ waitList: MutableList<String>
+ occupacy: Int
+ numWinners: Int
+ registrationStart: Long
+ registrationEnd: Long
+ bookingStart: Long
+ bookingEnd: Long
+ associatedSites: MutableList<String>
+ lotteryID: String
+ lotteryOwnerID: String
+ winners: MutableLIst<String>
+ losers: MutableList<String>
+ addWinner(id: String):Unit
+ addLoser(id: String): Unit
Reservation
+ Dates: List<Long?>()
Campsite
+ name: String
+ ID: String
+ location: Location
+ locationName: String
+ tags: MutableList<String>
+ price: Float
+ rating: Float?
+ rules: String
+ description: String
+ occupancy: Int
+ images: MutableList<Blob>
+ ownerID: String
+ availability: MutableList<Long>
CampsiteList
+ addCampsite(site: Campsite): Unit
+ updateCampsite(oldSite: Campsite, updatedSite: Campsite): Unit
+ deleteCampsite(site: Campsite): Unit
+ getOwnerSites(ID: String): List<Campsite>
+ getCampsite(ID: String): Unit
OwnerList
+ addOwner(owner: Owner): Unit
+ updateOwner(owner: Owner): Unit
+ deleteOwner(owner: Owner): Unit
+ getOwner(id: String): Unit
AdminList
+ addAdmin(admin: Admin): Unit
+ updateAdmin(admin: Admin): Unit
+ deleteAdmin(admin: Admin): Unit
+ getAdmin(id: String): Unit
MainActivity
BallinApp
+ NavHost(navController = NavController, StartDestination = “Signup”)
CampsiteListScreen
+ CampsiteListScreen(
navController: NavController,
ownerID: String,
getOwnerSites:() → List<Campsite>
): Unit
+ CampsiteItem(site:Campsite, onClick:()->Unit):Unit
CampsiteDetailScreen
+ CampsiteDetailScreen(
navController: NavController,
site:Campsite,
userID: String
): Unit
+ ImageDisplayRow(image: Blob):Unit
MainOwnerMenu
+OwnerMenuScreen(navController: NavController, ownerID: String)
+OwnerOptionsButton(
name: String,
destination: String,
testTag: String,
navController: NavController)
CampsiteInputScreen
+ CampsiteInputScreen(
navController : NavController,
addsite: (Campsite) → Unit,
updateSite: (oldsite: Campsite, newSite: Campsite) → Unit)
+ ImageRow(image: Blob, onDelete: (Blob)->Unit)
+TagRow(tag: String, onDelete:() → Unit)
+ resizePhoto(bitmap: Bitmap) → Bitmap
CampsiteActiveReservations
LotteryListScreen
+ LotteryListScreen(navController: NavController, ownerID: String,
getOwnerLotteries: (String) -> List<Lottery>,
getCampsite: (String) -> Campsite?) : Unit
+ LotteryListView(navController: NavController, ownerID: String,
lotteries: List<Lottery>, getCampsite: (String) -> Campsite?
) : Unit
+ NewLotteryButton(navController: NavController, ownerID: String
) : Unit
+ LotteryItem(lottery: Lottery, onClick: () -> Unit,
getCampsite: (String) -> Campsite?) : Unit
+ formatDateMillisToDMY(millis: Long): String
LotteryDetailScreen
+ LotteryDetailScreen(navController: NavController,
lottery: Lottery?, ownerID: String, ownerSites: List<Campsite>,
getCampsite: (String) -> Campsite?,
addLottery: (Lottery) -> Unit,
updateLottery: (oldLottery: Lottery, newLottery: Lottery) -> Unit
) : Unit
+ CampsiteSelectable(campsite: Campsite, isSelected: Boolean,
onClick: () -> Unit) : Unit
+ formatDateMillisToDMY(millis: Long): String
AdminMenuScreen
- navController: NavController
+ AdminMenuScreen(navController)
+ AdminOptionButton(name: String, destination: String, testTag: String, navController)
AdminCampsitesListScreen
- navController: NavController
- campsites: List<Campsite>
- siteToDelete: Campsite?
- query: String
- results: List<Campsite>
+ AdminCampsiteListScreen(navController, campsites, onDelete: (Campsite) -> Unit)
+ AdminCampsiteListView(campsites, onCampsiteClick: (Campsite) -> Unit, onDelete: (Campsite) -> Unit)
+ AdminCampsiteItem(site: Campsite, onClick: () -> Unit, onRemoveClick: () -> Unit)
+ formatPrice(price: Float): String
AdminCampsiteDetailScreen
- navController: NavController
- site: Campsite?
- showDialog: Boolean
+ AdminCampsiteDetailScreen(navController, site, onDelete: (Campsite) -> Unit)
+ AdminCampsiteDetailView(site: Campsite?, onDelete: (Campsite) -> Unit)
+ CampsiteDetailRow(label: String, value: String)
+ CampsiteImageGallery(images: List<Blob>)
+ CampsiteImage(image: Blob, index: Int)
AdminProfileListScreen
- navController: NavController
- profiles: List<Profile>
- profileToDelete: Profile?
- results: List<Profile>
+ AdminProfileListScreen(navController, profiles, onDelete: (Profile) -> Unit)
+ AdminProfileListView(profiles, onProfileClick: (Profile) -> Unit, onDelete: (Profile) -> Unit)
+ ProfileItem(profile: Profile, onClick: () -> Unit, onRemoveClick: () -> Unit)
AdminProfileDetailScreen
- navController: NavController
- profile: Profile?
- showDialog: Boolean
+ AdminProfileDetailScreen(navController, profile, onDelete: (Profile) -> Unit)
+ AdminProfileDetailView(profile: Profile?, onDelete: (Profile) -> Unit)
+ ProfileDetailRow(label: String, value: String)
MessagesListScreen
+MessageList(navController: NavController, messages: List<Message>)
+MessageButtonRow(index: Int, message: Message, navController: NavController)
MessageDetailScreen
+MessageDetail(navController: NavController, message: Message, onDelete: (Message) → Unit)
+<T>MessageDetailRow(descriptionText: String, content T)
NotificationListScreen
+ NotificationList(navController: NavController, notificationList: NotificationList)
+NotificationButtonRow(index: Int, notification: Notification, navController: NavController)
NotificationDetailScreen
+ NotificationDetail(navController: NavController, notification: Notification): Unit
+<T>NotificationDetailRow(descriptionText: String, content T)
MainEntrantMenu
+ navController: NavController
EntrantBrowseReservations
+ BrowseReservations(navController: NavController, campsiteList: CampsiteList, entrant: Entrant): Unit
ActiveReservationScreen
+ ActiveReservations(navController: NavController, entrant: Entrant, campsiteList: CampsiteList): Unit
PastReservationScreen
+ PastReservations(navController: NavController, entrant: Entrant): Unit
EntrantSiteBooker
+ SiteBooker(navController: NavController, campsiteID: String, campsiteList: CampsiteList, entrant: Entrant): Unit
EntrantBrowseCampsites
+ navController: NavController
+ campsiteList: CampsiteList
+ entrantID: String
PolicyGuidelinesScreen
+ PolicyGuidelines(navController: NavController): Unit
QRCodeScanner
+ navController: NavController
Uses
SearchBar
+guestFilter(view_model: SearchViewModel)
+locationFilter(view_model: SearchViewModel
+DisplayCampsite(NavController)
+Searchbar(viewModel: SearchViewModel = viewModel())
Uses
SearchViewModel
-_query: mutableStateFlow()
+query: _query.asStateFlow()
-_selectedNumofGuests(): MutableStateFlow<Int?>(null)
+ numOfGuests: _selectedNumOfGuests.asStateFlow()
-_selectedLocationFilter = MutableStateFlow<String?>(null)
+locationFilter = _selectedLocationFilter.asStateFlow()
-_selecteddates //to be added
-_campsites = MutableStateFlow<List<Campsite>>(emptyList())
+onSearchTextChange(text: String)
+onSelectedLocationFilterChange(locationFilter: String?)
+onNumOfGuestsChange(dateFilter: String)
+updateCampsites(newList: List<Campsite>)

1